



Name: _____

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Score: / 10

Learning Objectives

- Apply the Product Rule to differentiate products of functions
- Apply the Quotient Rule to differentiate rational functions
- Use the Chain Rule for composite functions
- Differentiate implicitly and find second derivatives

Identify which rule applies before differentiating. For the Chain Rule, explicitly name the outer and inner functions.

1. Differentiate using the Product Rule.

$$f(x) = x^3 \cdot e^x$$

Answer: _____

2. Differentiate using the Product Rule.

$$f(x) = (2x^2 + 1)\sin x$$

Answer: _____

3. Differentiate using the Quotient Rule.

$$f(x) = \frac{x^2 + 1}{x - 3}$$

Answer: _____

4. Differentiate using the Quotient Rule.

$$f(x) = \frac{\sin x}{x^2}$$

Answer: _____

5. Differentiate using the Chain Rule.

$$f(x) = (3x^2 - 5)^6$$

Answer: _____

6. Differentiate using the Chain Rule.

$$f(x) = e^{x^2 + 2x}$$

Answer: _____



7. Differentiate using the Chain Rule.

$$f(x) = \sin(3x^2)$$

Answer: _____

8. Differentiate using the Chain Rule with a natural log.

$$f(x) = \ln(x^2 + 4)$$

Answer: _____

9. Find $f''(x)$. (Second derivative)

$$f(x) = x^5 - 4x^3 + 2x$$

Answer: _____

10. Use implicit differentiation to find dy/dx .

$$x^2 + y^2 = 25$$

Answer: _____





Problems 5–8 cover Chain Rule in different contexts (power, exp, trig, log). Problem 10 (implicit differentiation) — remind students to treat y as a function of x .

Solutions

1. Differentiate using the Product Rule.

$$f(x) = x^3 \cdot e^x$$

→ Product Rule: $(uv)' = u'v + uv'$.

→ $u = x^3$, $u' = 3x^2$. $v = e^x$, $v' = e^x$.

→ $f'(x) = 3x^2 \cdot e^x + x^3 \cdot e^x = e^x(x^3 + 3x^2)$.

Answer: $f'(x) = x^3 e^x + 3x^2 e^x = e^x(x^3 + 3x^2)$

2. Differentiate using the Product Rule.

$$f(x) = (2x^2 + 1)\sin x$$

→ $u = 2x^2 + 1$, $u' = 4x$. $v = \sin x$, $v' = \cos x$.

→ $f'(x) = 4x \sin x + (2x^2 + 1) \cos x$.

Answer: $f'(x) = 4x \sin x + (2x^2 + 1) \cos x$

3. Differentiate using the Quotient Rule.

$$f(x) = \frac{x^2 + 1}{x - 3}$$

→ Quotient Rule: $(u/v)' = (u'v - uv') / v^2$.

→ $u = x^2 + 1$, $u' = 2x$. $v = x - 3$, $v' = 1$.

→ Numerator: $2x(x - 3) - (x^2 + 1) = 2x^2 - 6x - x^2 - 1 = x^2 - 6x - 1$.

→ $f'(x) = (x^2 - 6x - 1) / (x - 3)^2$.

Answer: $f'(x) = \frac{(x - 3)(2x) - (x^2 + 1)}{(x - 3)^2} = \frac{x^2 - 6x - 1}{(x - 3)^2}$

4. Differentiate using the Quotient Rule.

$$f(x) = \frac{\sin x}{x^2}$$

→ $u = \sin x$, $u' = \cos x$. $v = x^2$, $v' = 2x$.

→ Numerator: $x^2 \cdot \cos x - \sin x \cdot 2x = x(x \cos x - 2 \sin x)$.

→ $f'(x) = (x \cos x - 2 \sin x) / x^3$.

Answer: $f'(x) = \frac{x \cos x - 2 \sin x}{x^3}$



5. Differentiate using the Chain Rule.

$$f(x) = (3x^2 - 5)^6$$

→ Chain Rule: $d/dx[g(x)^n] = n \cdot g(x)^{n-1} \cdot g'(x)$.

→ Outer: $6(3x^2 - 5)^5$. Inner derivative: $6x$.

→ $f'(x) = 36x(3x^2 - 5)^5$.

Answer: $f'(x) = 6(3x^2 - 5)^5 \cdot 6x = 36x(3x^2 - 5)^5$

6. Differentiate using the Chain Rule.

$$f(x) = e^{x^2 + 2x}$$

→ $d/dx[e^u] = e^u \cdot u'$, where $u = x^2 + 2x$.

→ $u' = 2x + 2$.

→ $f'(x) = (2x + 2)e^{x^2 + 2x}$.

Answer: $f'(x) = (2x + 2)e^{x^2 + 2x}$

7. Differentiate using the Chain Rule.

$$f(x) = \sin(3x^2)$$

→ $d/dx[\sin(u)] = \cos(u) \cdot u'$, where $u = 3x^2$.

→ $u' = 6x$.

→ $f'(x) = 6x \cos(3x^2)$.

Answer: $f'(x) = \cos(3x^2) \cdot 6x$

8. Differentiate using the Chain Rule with a natural log.

$$f(x) = \ln(x^2 + 4)$$

→ $d/dx[\ln u] = u'/u$, where $u = x^2 + 4$.

→ $u' = 2x$.

→ $f'(x) = 2x / (x^2 + 4)$.

Answer: $f'(x) = \frac{2x}{x^2 + 4}$

9. Find $f''(x)$. (Second derivative)

$$f(x) = x^5 - 4x^3 + 2x$$

→ $f'(x) = 5x^4 - 12x^2 + 2$.

→ $f''(x) = 20x^3 - 24x$.

Answer: $f''(x) = 20x^3 - 24x$

10. Use implicit differentiation to find dy/dx .

$$x^2 + y^2 = 25$$

→ Differentiate both sides with respect to x .

→ $2x + 2y \cdot (dy/dx) = 0$.

→ Solve: $dy/dx = -x/y$.

Answer: $\frac{dy}{dx} = -\frac{x}{y}$

